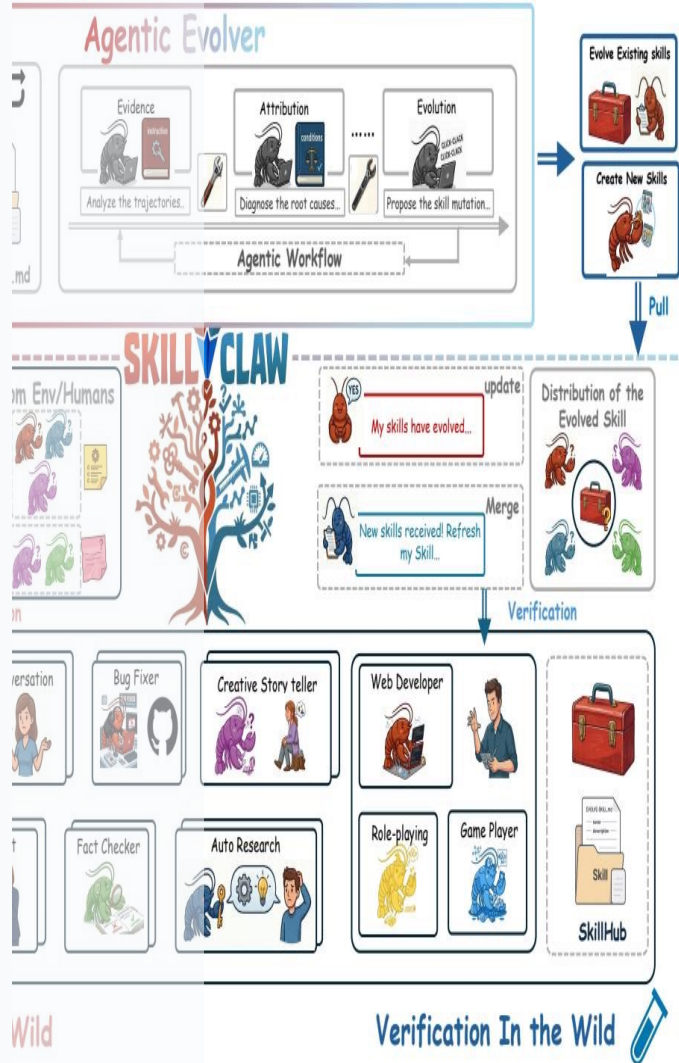


SkillClaw

Let Skills Evolve Collectively with Agentic Evolver

Collective Skill Evolution for Multi-User LLM Agent
Ecosystems

arxiv.org/abs/2604.08377



Agent Skills Remain Static After Deployment — Users Rediscover Solutions Independently

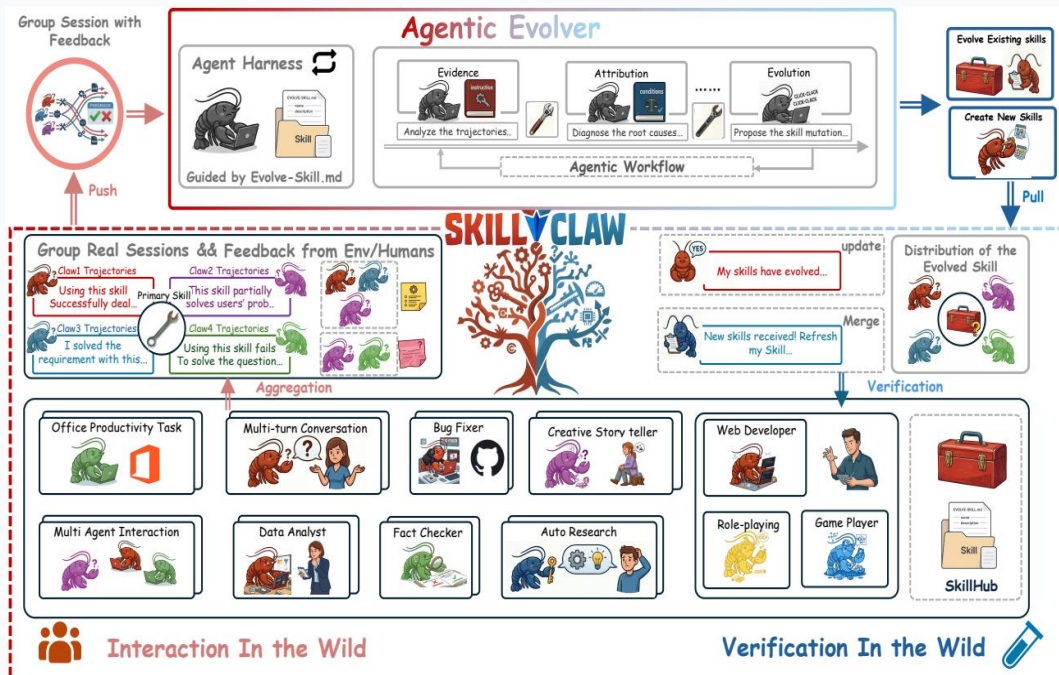
The Problem with Static Skills

- Skills are manually installed from a centralized hub and never updated from runtime experience.
- Similar tasks recur across users, but failures and solutions are repeatedly rediscovered.
- The system fails to improve over time, preventing knowledge accumulation.

Limitations of Existing Paradigms

- Memory-based methods: Store past trajectories but remain tied to specific instances and fail to generalize.
- Skill-based methods: Compress experience into structured instructions but treat the library as a static resource.

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



Four Stages of Evolution

1. Multi-User Interaction

Generating structured session trajectories

2. Session Collection

Aggregating evidence across the ecosystem

3. Agentic Evolver

Analyzing, attributing, and evolving skills

4. Skill Synchronization

Distributing verified skills back

Agentic Evolver: Evidence → Attribution → Evolution

in Three Stages

1. Evidence

Analyze aggregated trajectories to identify robust behavioral patterns, success modes, and common failure paths across users.



2. Attribution

Diagnose root causes of observed failures and successes. Agentic reasoning isolates why a skill fails in specific contexts.



3. Evolution

Propose concrete skill mutations—refining existing skills or creating new ones. Driven by open-ended reasoning, not predefined rules.

Crucially, this relies on autonomous agent open-ended reasoning over evidence.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

The Power of Ecosystem Aggregation

- Compatible Systems: OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw.
- Diverse Contexts: Different users exercising the same skill produce complementary signals.

Why Single-User Signals Are Insufficient

- A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.
- Aggregation reveals both conditions under which a skill works and where it breaks, mapping its behavioral boundary.

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution

Day	Evolution Attempt / Candidate Skill	Outcome	Rationale
1	validate-tmp-workspace-inputs	Accepted	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Rejected	Keep current best pool
3	multimodal-creative-task-pipeline	Rejected	Keep current best pool
4	creative-pipeline (impr: img/visuals)	Rejected	Keep current best pool
5	creative-pipeline + fail-fast validation	Rejected	Keep current best pool
6	creative-pipeline (extended doc paths)	Rejected	Not admitted to next cycle

Conservative Verification: Only 1 of 6 candidates was accepted, preventing regressions.

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Day	Evolution Attempt Summary	Outcome	Rationale
1	git-push-with-auth-fallback (safe fallback)	Accepted	Add to Safety best pool
2	Unified patch filenames, reduced inconsistency	Accepted	Keep updated best pool
3	Auth-alt paths & secrets audit; fixed pitfalls	Accepted	Keep current best pool
4	Same-pool retest (validator confirmed current)	Accepted	Continue same best pool
5	Added push hang handled as auth failure	Rejected	No improvement
6	Added identity config & filename consistency	Rejected	Not admitted

Iterative Success: 3 of 6 days yielded accepted updates, showing successful iterative refinement.

Key Takeaways: Three Properties That Distinguish SkillClaw

1. Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem.

2. Fully Automatic

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention.

3. Agentic Evolution Paradigm

Skill updates are produced through open-ended reasoning rather than predefined update rules.

Implications